MILITARY SPECIFICATION

SPIN SYSTEM, ATOMIC WARHEAD SECTION: M30

This specification is approved for use by Picatinny Arsenal, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of control system used to impart a stabilizing spin to the Pershing warhead section, during flight, and is designated as Spin System, Atomic Warhead Section: M30.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-I-6866 - Inspection, Penetrant Method of.
* MIL-C-50963 - Cables, Warhead Section.

STANDARDS

FEDERAL

* FED. STD. No. 356 - Commercial Packaging of Supplies and Equipment.

FSC 1115

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MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

* MIL-STD-1460 - Soldering of Electrical Connections and Printed Wiring Assemblies, Procedure for.

DRAWINGS

* ARMY BALLISTIC MISSILE AGENCY

10560444 - Nozzle, Spin System.
10560725 - Gas Generator Assembly.
10560726 - Manifold, Machined.
10560740 - Shorting Plug Assembly.
10560742 - Case (gas generator).
10560743 - Igniter Assembly.
10560750 - Manifold Assembly.
10560759 - Plug, Burst Assembly.
10561076 - Spin System, Installation of.
10561083 - Connector Switch Assembly.
10564664 - Tube Assembly, Metal.
10564667 - Tube Assembly, Metal.
10564669 - Bracket, Mounting.
10564693 - Connector Switch.
10565443 - Connector Switch.
10623667 - Bracket, Angle.

PICATINNY ARSENAL

8861961 - Drum, Metal, Shipping, for Generator, Gas Pressure.
8864799 - Spin System, Atomic Warhead Section: M30.

PUBLICATIONS

PICATINNY ARSENAL

ET-8864799 - Inspection Equipment Tabulation.
QAI-PA-1 - Quality Assurance Instructions for Reporting Data on Serialized Items.

Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.

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3. REQUIREMENTS

3.1 Materials and components.- Materials shall conform to applicable specifications and drawings. Each component shall conform to all dimensions and tests specified on the component drawing. The contractor shall have available verifiable proof (i.e., objective evidence (see 6.2)) that materials and components were fabricated, inspected, and tested under controllable conditions as set forth in the contractor's quality control or inspection procedures.

3.2 Assembly.- The spin system shall comply with applicable requirements specified on drawing (dwg.) 8864799, with all requirements specified herein, and with the requirements of applicable specifications to the extent specified on the drawings and herein.

NOTE: For purposes of this specification, the term "spin system" shall mean an assembly consisting of two nozzles, two metal tube assemblies, a manifold assembly, three brackets, and a gas generator assembly (see dwg. 8864799).

* 3.2.1 Connector switches.- The connector switches, referenced on dwg. 8864799, shall be procured and shall comply with the specifications listed on dwg. 10564693 and 10565443 (covering a detail of dwg. 10561083).

* 3.2.2 Wiring Harness: 4W67.- The 4W67 wiring harness, referenced on dwg. 8864799, shall be procured and shall comply with MIL-C-50963.

3.3 First article.- When specified, the contractor shall furnish sample units for first article inspection and approval (see 4.3 and 6.1).

3.4 Gas generator assembly tests (nonoperational).

3.4.1 Shorting plug continuity.- There shall be continuity between adjacent sockets of the shorting plug assembly. Testing shall be as specified in 4.5.1.1.

3.4.2 Continuity and bridgewire resistance.- The continuity and bridgewire resistance of the igniter assembly shall be as specified in Table I. In addition, the resistance values specified with a code number shall be recorded (see 3.9.1). Testing shall be as specified in 4.5.1.2.

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TABLE I

From pin	To pin	Resistance (ohms)	Test Identification Number (see 3.9.1) A/
A	B	0.95 ± 0.25	042001
B	C	0.95 ± 0.25	042002
D	E	0.10 max.	--

A/ Not required during the following tests: High temperature, low temperature, altitude, temperature shock, humidity, shock, and vibration.

3.4.3 Insulation resistance.- The insulation resistance of the igniter assembly shall be not less than 2.0 megohms. The test potential shall be applied between mutually isolated pins and between each isolated pin(s) and igniter body (for a total of six test points). There shall be no evidence of insulation breakdown or other damage during this test. Testing shall be as specified in 4.5.1.3.

3.4.4 No-fire current.- The igniter assembly shall not fire when a direct current of 0.50 +0 -0.05 ampere is allowed to flow through each of the two bridgewires. The current shall flow through each bridgewire for a mimimum period of 5 minutes. Testing shall be as specified in 4.5.1.4.

3.5 Gas generator assembly tests (operational).- These tests constitute the firing of the spin system (see 4.5.2). The spin rate calculation of the spin system (see 3.8.9) follows these tests.

3.5.1 Ignition (fire) current.- The gas generator shall fire when a direct current of 2.0 +0 -0.5 ampere is allowed to flow through either one of the two bridgewires. The current shall flow through the bridgewire for a maximum period of 25 milliseconds. Testing shall be as specified in 4.5.2.1.

3.5.2 Ignition delay.- Ignition of the propellant shall occur not more than 100 milliseconds after application of the fire signal. Testing shall be as specified in 4.5.2.2.

3.5.3 Burning time.- The burning of the propellant shall be completed between 120 to 400 milliseconds. Testing shall be as specified in 4.5.2.3.

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3.6 Autoignition temperature.

3.6.1 Igniter assembly.- The igniter assembly shall not ignite while subjected to a temperature of 300° ± 5°F for a minimum period of 1 hour. Testing shall be as specified in 4.5.3.

3.6.2 Gas generator assembly.- The gas generator assembly shall not ignite while subjected to a temperature of 275° ± 5°F for a minimum period of 1 hour. Testing shall be as specified in 4.5.3.

3.7 Component proof-pressure tests.

* 3.7.1 Nozzle.- The nozzle shall withstand a proof pressure of 1000 ± 50 pounds per square inch gage (psig) for a minimum period of 20 seconds, and there shall be no evidence of cracking, swelling, leakage, or other damage. During this 20-second test period, a maximum of five pin-hole leaks is permitted providing the pressure drop does not exceed 100 pounds per square inch (psi). Testing shall be as specified in 4.5.4.

3.7.2 Burst plug assembly.- The burst plug assembly shall withstand a proof pressure of 1600 ± 50 psig for a minimum period of 20 seconds, and there shall be no evidence of rupture, leakage, pressure drop, abnormal distortion, or other damage. Testing shall be as specified in 4.5.4.

3.7.3 Case (gas generator).- The gas generator case shall withstand a proof pressure of 3200 ± 10 psig for a period of 60 ± 5 seconds, and there shall be no evidence of cracking, swelling, distortion, leakage, pressure drop, or other damage. Testing shall be as specified in 4.5.4.

3.7.4 Machined manifold.- The machined manifold (with a plug installed in the burst plug assembly hole) shall withstand a proof pressure of 3200 ± 10 psig for a minimum period of 60 seconds, and there shall be no evidence of cracking, distortion, leakage, pressure drop, or other damage. Testing shall be as specified in 4.5.4.

3.7.5 Tube assembly.- The tube assembly shall withstand a proof pressure of 3000 ± 25 psig for a minimum period of 20 seconds, and the leakage rate shall not exceed 5 psi per minute. There shall be no evidence of cracking, swelling, distortion, or other damage. Testing shall be as specified in 4.5.4.

* 3.7.6 Igniter assembly (prior to assembling bridgewires, potting compound, and detonating charges).- The glass-to-metal seals of the igniter assembly shall withstand a proof pressure of 5000 ± 50 psi absolute for a minimum period of 5 seconds,

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and there shall be no evidence of leakage, pressure drop, cracking, or other damage during this test. Testing shall be as specified in 4.5.4.

3.8 Spin system tests.

3.8.1 Burst disc rupture test.- The burst disc rupture pressure shall be between 2100 to 3000 psig. Testing shall be as specified in 4.5.5.1.

3.8.2 High temperature.- The spin system shall withstand a high temperature of 160° ± 5°F, for a minimum period of 6 hours, without evidence of corrosion, deterioration, or damage that may impair its intended operation. The unit to be fired (see Table III) shall also comply with the requirements of 3.4.2, 3.4.3, 3.5.1, 3.5.2, 3.5.3, and 3.8.9 while at the high temperature (at the end of the high-temperature exposure period). Testing shall be as specified in 4.5.5.2.

3.8.3 Low temperature.- The spin system shall withstand a low temperature of -33° ± 3°F for a minimum period of 6 hours, without evidence of corrosion, deterioration, or damage that may impair its intended operation. The unit to be fired (see Table III) shall also comply with the requirements of 3.4.2, 3.4.3, 3.5.1, 3.5.2, 3.5.3, and 3.8.9 while at the low temperature (at the end of the low-temperature exposure period). Testing shall be as specified in 4.5.5.3.

3.8.4 Altitude.- The spin system shall withstand a reduced atmospheric pressure corresponding to an altitude of 100,000 ± 2000 feet without evidence of deterioration or damage that may impair its intended operation. The unit to be fired (see Table III) shall also comply with the requirements of 3.4.2, 3.4.3, 3.5.1, 3.5.2, 3.5.3, and 3.8.9 while at the reduced test pressure. Testing shall be as specified in 4.5.5.4.

3.8.5 Temperature shock.- The spin system shall withstand alternate exposures to temperatures of 160° ± 5°F and -65° ± 5°F, for a minimum period of 4 hours at each exposure, and there shall be no evidence of corrosion, deterioration, or damage that may impair its intended operation. The unit to be fired (see Table III) shall also comply with the requirements of 3.4.2, 3.4.3, 3.4.4, 3.5.1, 3.5.2, 3.5.3, and 3.8.9 while at room temperature (at the end of the last low-temperature exposure period). Testing shall be as specified in 4.5.5.5.

* 3.8.6 Humidity.- The spin system shall withstand a maximum relative humidity of 95 ± 5 percent and a maximum temperature of 160° ± 5°F without evidence of corrosion, deterioration, or damage that may impair its intended operation. The unit to be fired (see Table III) shall also comply with the requirements

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of 3.4.2, 3.4.3, 3.4.4, 3.5.1, 3.5.2, 3.5.3, and 3.8.9 after humidity exposure. Testing shall be as specified in 4.5.5.6.

3.8.7 Shock.- The spin system shall withstand shocks of 20 ± 2 gravities (g) peak amplitude (sinusoidal) with a shock duration of 11 ± 1 millisecond as measured between points which are 10 percent of the shock amplitude. There shall be no evidence of deterioration or damage that may impair its intended operation. The unit to be fired (see Table III) shall also comply with the requirements of 3.4.2, 3.4.3, 3.4.4, 3.5.1, 3.5.2, 3.5.3, and 3.8.9 after shock testing. Testing shall be as specified in 4.5.5.7.

3.8.8 Vibration.- The spin system shall withstand vibrations as specified in Table II without evidence of deterioration or damage that may impair its intended operation. The unit to be fired (see Table III) shall also comply with the requirements of 3.4.2, 3.4.3, 3.4.4, 3.5.1, 3.5.2, 3.5.3, and 3.8.9 after vibration testing. Testing shall be as specified in 4.5.5.8.

TABLE II

Frequency Range (hertz)	Displacement (D.A.) or Peak g Level (± 10% Tol.)	Minimum Time Period to Cover Freq. Range (minutes)
10-30	0.20 inch	5
30-1000	8 g	5
1000-2000	10 g	5

3.8.9 Spin rate (calculated).- The spin system shall produce a calculated spin rate of 35 ± 6 revolutions per minute when attached to a test fixture having a roll moment of inertia of 9.0 slug-feet^2. Testing and spin rate calculations shall be as specified in 4.5.5.9.

* 3.8.10 Weight.- The weight of the spin system shall be no greater than 3.7 pounds. Testing shall be as specified in 4.5.5.10.

3.9 Serialization.

* 3.9.1 Gas generator assembly.- A six-letter serial code shall be assigned to each serialized production item. The contractor shall obtain these serial codes by means of a letter addressed to: Commander, Picatinny Arsenal, Dover, New Jersey 07801, Attention: SARPA-QA-N-A. The letter of request shall include:

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(a) exact nomenclature of the item as specified on the applicable drawing (see dwg. 10560725), (b) contract number or procurement request order number, and (c) number of serial codes required (unused codes must be returned). In addition to the normal inspection records, the contractor shall provide data feedback (see 6.1.2) in conformance with Quality Assurance Instruction QAI-PA-1. The data required to be coded, and the test identification numbers, are specified in 3.4.2.

* 3.9.2 Spin system components.- A serial number shall be assigned by the contractor to each spin system component when required by the applicable drawing, and in no event shall serial numbers be duplicated. Six-letter serial codes shall not be used on these components. If no serial number is required by the applicable drawing, the contractor may, at his own discretion, use serialization for purposes of internal control.

* 3.10 Soldering.- All soldering shall be in accordance with MIL-STD-1460.

3.11 Workmanship.- All parts and accessories shall be constructed, assembled, and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness, marking, cleaning, and freedom of the parts from burrs and sharp edges which may affect performance or result in injury during handling. All parts shall be free of chips, dirt, grease, rust, and other foreign material. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

* 4.2 Classification of inspections.- The inspection requirements specified herein are classified as follows:
    1. First Article Inspection (see 4.3)
    2. Quality Conformance Inspection (see 4.4)

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4.3 First article inspection.

* 4.3.1 Submission.- When the contract requires first article inspection, the contractor shall submit the following:
  (a) Nine disassembled spin systems (see 3.2 (Note)).
  (b) Two separate igniter assemblies (dwg. 10560743) (for the autoignition temperature test).
  (c) Four separate igniter assemblies (dwg. 10560743, prior to assembling bridgewires, potting compound, and detonating charges) (for the proof-pressure test).
  (d) Four separate gas generator cases (dwg. 10560742) (For defect inspection (see 4.4.2.1.1) and for the proof-pressure test).

* 4.3.2 Inspection and testing to be performed.- The first article inspection samples shall be inspected for the defects specified in 4.4.2.1 (except 4.4.2.1.10, 4.4.2.1.11, and 4.4.2.1.12), using inspection methods contained therein, and for all requirements of Section 3 (also see Table III) using the test methods specified in 4.5. Inspection and testing shall be performed with the acceptance inspection equipment specified in 4.4.4. The spin systems shall be tested in the order specified in Table III. For tests 8 to 12, the nine spin systems shall be assembled in their proper configuration by using the impulse table test fixture (see 4.5), or an approved equivalent, for assembly purposes. After being assembled, each spin system shall be removed from the assembly fixture and left in its assembled configuration for testing. If any spin system, or component, fails to comply with any of the applicable requirements, the first article samples shall be rejected. This series of tests (with the exception of test 1 and 2) is considered destructive.

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* TABLE III. First article tests.

Test	Requirement paragraph or dwg.	No. of units to be tested
1. Weight	3.8.10	9
2. Penetrant test of mounting bracket	dwg. 10564669	18
3. Component proof-pressure tests	3.7	9 A/
4. Shorting plug continuity	3.4.1	9
5. Continuity and bridgewire resistance	3.4.2	9
6. Insulation resistance	3.4.3	9
7. No-fire current	3.4.4	9
8. High temperature	3.8.2	9 B/
9. Low temperature	3.8.3	8 B/
10. Altitude	3.8.4	7 B/
11. Temperature shock	3.8.5	6 B/
12. Humidity	3.8.6	5 B/
13. Shock	3.8.7	4 B/
14. Vibration	3.8.8	3 B/
15. Autoignition temperature	3.6.1 C/ & 3.6.2	1 D/
16. Burst disc rupture test	3.8.1	1 D/

A/ Nine sets of spin system components shall be subjected to this test with the exception of the gas generator case and the igniter assembly. The four separate gas generator cases and the four separate igniter assemblies, submitted as part of the first article samples, shall be used for this test (see 4.3.1(c) and 4.3.1(d)).

B/ One of these spin systems shall be selected and assembled to the impulse table test fixture (see 4.5) before being subjected to this test, and it shall be fired after this test (see 3.5). For test 13 and 14, each spin system shall be mounted to the test fixture for testing.

C/ The two separate igniter assemblies, submitted as part of the first article samples, shall be used for this test (see 4.3.1(b)).

D/ After test 14, there remain two complete spin systems to be tested. One of them shall be used for test 15 and the other shall be used for test 16.

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4.4 Quality conformance inspection.

* 4.4.1 Inspection lot.- A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot, and a batch shall mean an inspection batch for the purposes of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor shall be designated or approved by the Government representative.

4.4.2 Product inspection examination.- A sample shall be selected at random from each inspection lot in accordance with the major defect or the minor defect table as applicable (see Table IV). Any defect found in the sample, selected for the major or minor classification, shall reject the lot. Critical defects shall require 100 percent inspection of all items in the inspection lot. If any critical defect is found in an item, the item shall be removed from the lot. The sequence of inspections for the classification of defects need not be followed so long as all the defect inspections are performed.

* TABLE IV. Inspection lot sampling.

Major Defect Table		Minor Defect Table	
Lot Size	Sample Size	Lot Size	Sample Size
1-30	all	1-30	all
31-45	29	31-45	27
46-60	35	46-60	30
61-75	38	61-75	33
76-95	42	76-95	35
96-130	46	96-130	37
131-180	50	131-180	39
181-250	52	181-250	41
251-400	54	251-400	42
401 and over	56	401 and over	44

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* 4.4.2.1 Classification of defects.- Critical, major, and minor defects are defined in MIL-STD-105.

* 4.4.2.1.1 Case (gas generator) (see dwg. 10560742). A/

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:		
101.	Wall thickness, min	Gage
102.	Pitch diameter of larger thread	Gage
103.	Case damaged (including dents, punctures, and gouges)	Visual
Minor:		
201.	Width across hex flats, max	Gage
202.	Marking missing, improper, or illegible	Visual
203.	Protective finish missing or improper	Visual
204.	Evidence of poor workmanship (see 3.11)	Visual

A/ The four gas generator cases, submitted as part of the first article samples, shall be used for this inspection during first article testing.

* 4.4.2.1.2 Gas Generator Assembly (see dwg. 10560725).

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:		
101.	Weight	Approved weighing device
102.	Pin in igniter assembly missing or damaged (6 pins)	Visual
103.	Height of pins in igniter assembly (see 6.3)	Visual
104.	Shorting plug missing	Visual
105.	Case damaged (including dents, punctures, and gouges)	Visual
Minor:		
201.	Shipping plug missing	Visual
202.	Identification plate missing or improperly marked	Visual
203.	Protective finish missing or improper	Visual
204.	Evidence of poor workmanship (see 3.11)	Visual

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* 4.4.2.1.3 Manifold, Machined (see dwg. 10560726). A/

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:		
101.	Location of mounting hole (4 holes)	Gage
102.	Pitch diameter of thread on end port, max. (2 places)	Gage
103.	Major diameter of thread on end port, max. (2 places)	Gage
104.	Pitch diameter of large thread, max	Gage
105.	Major diameter of large thread, max	Gage
106.	Manifold cracked or damaged	Visual
Minor:		
201.	Internal configuration of end port (2 places)	Visual
202.	Internal configuration of internally threaded center port	Visual
203.	Evidence of poor workmanship (see 3.11)	Visual

A/ The burst plug assembly shall be removed from the manifold assembly (dwg. 10560750), for this inspection, during first article testing.

4.4.2.1.4 Manifold Assembly (see dwg. 10560750).

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:		
101.	Damaged threads (3 places)	Visual
102.	Manifold cracked or damaged	Visual
103.	Burst plug assembly missing, damaged, or improperly assembled	Visual
Minor:		
201.	O ring missing or damaged (inspect prior to assembly of burst plug)	Visual
202.	Marking missing, improper, or illegible	Visual
203.	Evidence of poor workmanship (see 3.11)	Visual

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4.4.2.1.5 Tube Assembly, Metal (see dwg. 10564664 and 10564667).

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:		
	101. Location of mounting hole in flange (4 holes)	Gage
	102. Concentricity of counterbore in flange with inside diameter of tube	Gage
	103. Depth of counterbore, min	Gage
	104. Radius of large bend in tube	Gage
	105. Radius of small bend in tube	Gage
	106. Parallelism of flange surface with tube (dwg. 10564664 only)	Gage
	107. Angular position of flange surface with tube (dwg. 10564664 only)	Gage
	108. Improper orientation of flange with tube	Visual
	109. Tube damaged (including dents, punctures, and gouges)	Visual
Minor:		
	201. Major diameter of O-ring groove in flange, max	Gage
	202. Fittings damaged or improperly installed	Visual/Manual
	203. Marking missing, improper, or illegible	Visual
	204. Evidence of poor workmanship (see 3.11)	Visual

* 4.4.2.1.6 Nozzle, Spin System (see dwg. 10560444).

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:		
	101. Location of mounting hole (4 holes)	Gage
	102. Length of insert extension, max	Gage
	103. Outside diameter of insert extension, max	Gage
	104. Throat diameter, max	Gage
	105. Diameter of nozzle outlet, max	Gage
	106. Height of nozzle inlet, max	Gage

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107. Smaller diameter of nozzle inlet, max.....Gage
108. Imperfections on exterior surface and on interior conical surface (including cracks, gouges, irregularities, voids, and exposed fibers).....Visual

Minor:
201. Insert missing or damaged.....Visual
202. Improper direction of insert mismatch.....Visual
203. Tape on nozzle outlet missing or improper.....Visual
204. Marking missing, improper, or illegible.....Visual
205. Evidence of poor workmanship (see 3.11).....Visual

4.4.2.1.7 Bracket, Mounting (see dwg. 10564669).

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:		
101.	Location of elongated mounting hole (3 holes)	Gage
102.	Location of round mounting hole (4 holes)	Gage
103.	Location of large hole in nozzle mounting surface	Gage
104.	Angular location of nozzle mounting surface	Gage
105.	Bracket improperly formed	Visual
106.	Welding cracked or improper	Visual
Minor:		
201.	Marking missing, improper, or illegible	Visual
202.	Protective finish missing or improper	Visual
203.	Evidence of poor workmanship (see 3.11)	Visual

* 4.4.2.1.8 Bracket, Angle (see dwg. 10623667).

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:		
101.	Location of mounting hole in larger surface (4 holes)	Gage

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	102.	Location of mounting hole in smaller surface (4 holes).....Gage
	103.	Angle of bend, min.....Gage
	104.	Objective evidence available for heat treatment (see 6.2).....Visual
Minor:		
	201.	Marking missing, improper, or illegible.....Visual
	202.	Protective finish missing or improper.....Visual
	203.	Evidence of poor workmanship (see 3.11).....Visual

* 4.4.2.1.9 Shorting Plug Assembly (see dwg. 10560740).

Categories		Defects	Method of Inspection
Critical:		None defined.	
Major:			
	101.	Soldering improper (inspect during assembly) (see 3.10).....Visual	A/
	102.	Socket missing or damaged (6 sockets).....Visual	
	103.	Connector damaged (including cracked shell or damaged insert material).....Visual	
Minor:			
	201.	Potting missing, damaged, or improper.....Visual	
	202.	Evidence of poor workmanship (see 3.11).....Visual	

A/ Not applicable for first article samples.

* 4.4.2.1.10 Drum, Metal, Shipping, for Generator, Gas Pressure (prior to securing cover) (see dwg. 8861961).

Categories		Defects	Method of Inspection
Critical:		None defined.	
Major:			
	101.	Gas generator improperly wrapped and bagged.....Visual/Manual	
	102.	Cushioning disk (2 places) and cushioning material improperly located or applied.....Visual/Manual	

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Minor:
201. Bag marking missing, improper, or illegible.....Visual
202. Evidence of poor workmanship (see 3.11).....Visual

* 4.4.2.1.11 Drum, Metal, Shipping, for Generator, Gas Pressure (see dwg. 8861961).

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:		
	101. Drum improperly sealed (including metallic seal).....	Visual/Manual
	102. Drum damaged (including punctures, dents, and gouges).....	Visual
Minor:		
	201. Scratch on drum exposing base metal.....	Visual
	202. Marking missing, improper, or illegible.....	Visual
	203. Evidence of poor workmanship (see 3.11).....	Visual

* 4.4.2.1.12 Packaging of Spin System Components (Excluding Gas Generator Assembly) (see 3.2 (Note)).

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:		
	101. Objective evidence available (see 6.2) that spin system components (excluding gas generator assembly) were preserved, packaged, packed, and marked in accordance with Section 5.....	Visual
Minor:	None defined.	

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4.4.3 Inspection testing.- The contractor shall perform all of the tests specified in Table V, and they shall be performed in the listed sequence. Of these tests, 8(a), 8(b), 8(c), and 9 are considered destructive, and samples so tested shall not be returned to the lot.

TABLE V. Quality conformance tests.

Test	Test Classification
1. Penetrant test	Minor
2. Weight	Minor
3. Component proof-pressure tests	Major
4. Shorting plug continuity	Minor
5. Continuity and bridgewire resistance	Major
6. Insulation resistance	Minor
7. No-fire current	Major
8. Gas generator assembly tests (operational):	
(a) Ignition current	Major
(b) Ignition delay	Major
(c) Burning time	Major
9. Spin rate	Major

4.4.3.1 Penetrant test on Bracket, Mounting (see dwg. 10564669).- Each mounting bracket shall be subjected to this test in accordance with MIL-I-6866. If any bracket fails to comply with the applicable requirements, it shall be classified defective and be removed from the lot.

* 4.4.3.2 Component proof-pressure tests (see 3.7) on Nozzle (dwg. 10560444); Burst Plug Assembly (dwg. 10560759); Case (gas generator) (dwg. 10560742); Manifold, Machined (dwg. 10560726); Tube Assembly, Metal (dwg. 10564664 and 10564667); and Igniter Assembly (dwg. 10560743).- Each component shall be subjected to this test (see 4.5.4). If any component fails to comply with the applicable requirements, it shall be classified defective and be removed from the lot.

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4.4.3.3 Shorting plug continuity (see 3.4.1), continuity and bridgewire resistance (see 3.4.2), insulation resistance (see 3.4.3), no-fire current (see 3.4.4), and weight (see 3.8.10).- Each shorting plug, igniter assembly, and spin system shall be subjected to these tests, as applicable (see applicable paragraph in 4.5). If any shorting plug, igniter assembly, or spin system fails to comply with the applicable requirements, it shall be classified defective and be removed from the lot.

4.4.3.4 Gas generator assembly tests (operational) (see 3.5): ignition current (see 3.5.1), ignition delay (see 3.5.2), and burning time (see 3.5.3). Spin rate (see 3.8.9).- One spin system shall be selected from every lot of 11 spin systems produced, or portion thereof, and subjected to these tests (see applicable paragraph in 4.5). If the sample fails to comply with the applicable requirements, the lot shall be rejected.

4.4.4 Acceptance inspection equipment.

4.4.4.1 Major defects and tests.- Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions as specified in the Equipment List (EL) listed in the Equipment Tabulation, ET-8864799. When either of the following conditions exists, the contractor shall design required equipment or specify suitable commercial equipment: (a) if any deviation from equipment or procedures, as specified in the EL, is desired or (b) the code "CD" is listed in the "Part or Identifying Number" column in the EL (see 6.1.2). The contractor shall obtain approval of such designs, operating instructions, and special calibration instructions prior to use in the contract (see 6.4).

4.4.4.2 Minor defects and tests.- Acceptance inspection equipment for minor defects is not specified in ELs. The acceptance inspection equipment, operating instructions, and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

* 4.5 Methods of inspection.- The tolerances specified in this specification are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used. The spin system to be fired shall be assembled to the impulse table test fixture as specified in EL-10561076 (see ET-8864799), and it shall be mounted in a similar manner as shown on dwg. 10561076 (except that the gas generator assembly shall be installed on the manifold assembly and torqued to 33 ± 3 foot-pounds). The spin system

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characteristics, during firing, shall be monitored by a recording oscillograph, pressure transducer, and other suitable measuring and recording equipment as required during the firing process. Unless otherwise specified, the tests shall be conducted at room temperature (77° ± 10°F). NOTE: During testing, the gas generator shall be connected to its shorting plug or to a mating connector.

4.5.1 Gas generator assembly tests (nonoperational).- These tests may be performed on a completely assembled spin system mounted to the test fixture specified in 4.5. The igniter assembly may be disconnected from the gas generator for tests on the igniter assembly (test 4.5.1.2, 4.5.1.3, and 4.5.1.4).

4.5.1.1 Shorting plug continuity.- Continuity between sockets of the shorting plug assembly shall be verified by use of a Government-approved continuity tester and shall comply with the requirements of 3.4.1 during this test.

4.5.1.2 Continuity and bridgewire resistance.- The continuity and bridgewire resistance of the igniter assembly shall be measured with a Government-approved tester. During bridgewire resistance measurements, the bridgewire current shall be limited to a maximum of 5 milliamperes. The igniter assembly shall comply with the requirements of 3.4.2 during this test.

* 4.5.1.3 Insulation resistance.- The insulation resistance of the igniter assembly shall be measured by using a Government-approved insulation resistance tester at an applied voltage of 500 ± 50 volts direct current. The test voltage shall be applied instantaneously, and the insulation resistance shall be measured after a 10-second application of this test voltage. The igniter assembly shall comply with the requirements of 3.4.3 during this test.

* 4.5.1.4 No-fire current.- The igniter bridgewire no-fire current shall be supplied by means of a Government-approved, direct-current power supply, and each bridgewire shall be tested separately. The igniter assembly shall comply with the requirements of 3.4.4 during this test.

4.5.2 Gas generator assembly tests (operational).- These tests shall be performed on a completely assembled spin system mounted to the test fixture specified in 4.5. These tests constitute the firing of the spin system.

4.5.2.1 Ignition (fire) current.- The igniter bridgewire fire current shall be supplied by means of a Government-approved, 28-volt direct-current power supply. The gas generator shall comply with the requirements of 3.5.1 during this test.

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4.5.2.2 Ignition delay.- Ignition delay time shall be measured by means of the recording equipment specified in 4.5. Ignition delay time shall be measured between the fire point and a point on the rise portion of the pressure curve equal to 10 percent of the maximum operating pressure. The maximum operating pressure shall be the highest pressure developed by the gas generator (excluding any pressure spike caused by ignition). The gas generator shall comply with the requirements of 3.5.2 during this test.

4.5.2.3 Burning time.- Propellant burning time shall be measured by means of the recording equipment specified in 4.5. Propellant burning time shall be measured between a point on the ascending portion of the pressure curve equal to 75 percent of the maximum operating pressure and a point on the descending portion of the pressure curve equal to 80 percent of the maximum operating pressure. The maximum operating pressure shall be the highest pressure developed by the gas generator (excluding any pressure spike caused by ignition). The gas generator shall comply with the requirements of 3.5.3 during this test.

* 4.5.3 Autoignition temperature.- The igniter assembly and the gas generator assembly shall be placed into suitable test chambers, and the temperature of each chamber shall be raised to the applicable test temperature. The igniter assembly and the gas generator assembly shall remain at their test temperatures for a minimum period of 1 hour, and they shall comply with the requirements of 3.6.1 or 3.6.2, as applicable.

* 4.5.4 Component proof-pressure tests.- Spin system components specified in 3.7 shall be subjected to hydrostatic pressure at the applicable specified level. During first article testing, the test on the gas generator case and on the igniter assembly shall be performed on the four separate gas generator cases and the four separate igniter assemblies submitted as part of the first article samples (see 4.3.1(c) and 4.3.1(d)). The burst plug assembly shall be removed from the nine manifold assemblies (dwg. 10560750), submitted for first article testing, for the test on the machined manifold and the burst plug assembly. Government-approved test setups, with suitable pressure monitoring devices, shall be used for these tests. Once the specified test pressure is attained, there shall be no further adjustment of the pressure during the test period. The components shall comply with the applicable requirements of 3.7 during these tests.

4.5.5 Spin system tests.

4.4.5.1 Burst disc rupture test.- A nozzle shall be removed from one of the tube assemblies and replaced with an adequate blocking plate. The other tube assembly shall be removed from

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the manifold assembly, and a suitable pressure transducer shall be installed in its place. The spin system shall then be fired, and burst pressure shall be measured by means of the recording equipment specified in 4.5. The burst disc shall comply with the requirements of 3.8.1 during this test.

4.5.5.2 High temperature.- The spin system shall be placed into a test chamber, and the temperature of the chamber shall be raised to 160°F. The spin system shall remain at this temperature for a minimum period of 6 hours. At the completion of this exposure period, the spin system shall comply with the requirements of 3.8.2.

4.5.5.3 Low temperature.- The spin system shall be placed into a test chamber, and the temperature of the chamber shall be lowered to -33°F. The spin system shall remain at this temperature for a minimum period of 6 hours. At the completion of this exposure period, the spin system shall comply with the requirements of 3.8.3.

4.5.5.4 Altitude.- The spin system shall be placed into an altitude chamber, and the internal pressure of the chamber shall be reduced to a pressure equivalent to an altitude of 100,000 ± 2000 feet. The spin system shall comply with the requirements of 3.8.4 while at this reduced pressure.

* 4.5.5.5 Temperature shock.- The spin system shall be placed into a test chamber having an internal temperature of 160°F, and it shall remain at this temperature for a minimum period of 4 hours. At the end of this test exposure, and within a 10-minute period, the spin system shall be transferred to a test chamber having an internal temperature of -65°F, and it shall remain at this temperature for a minimum period of 4 hours. This high- and low-temperature exposure constitutes one cycle, and the spin system shall be subjected to three such cycles. At the completion of the third cycle (at the end of the low-temperature exposure period), the spin system shall be removed from the test chamber and allowed to stabilize at room temperature. The spin system shall then comply with the requirements of 3.8.5.

NOTE: The test may, for convenience purposes, be interrupted at the end of a low-temperature exposure period. This is accomplished by returning the spin system to room temperature after an exposure at the high temperature of 1 hour minimum. To continue the test, start with the high-temperature exposure of the test cycle.

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* 4.5.5.6 Humidity.- The spin system shall be placed into a dry oven and conditioned at a temperature of 160°F for a minimum period of 24 hours. At the end of this conditioning period, the spin system shall be transferred to a humidity chamber having a temperature of 75° ± 5°F and a relative humidity of 50 ± 10 percent. The spin system shall remain in this chamber for a minimum period of 24 hours. At the end of this exposure period, the temperature of the chamber shall be gradually raised, within a 2-hour period, to 105° ± 5°F and a relative humidity of 95 percent. The spin system shall remain in this chamber for a minimum period of 16 hours. At the end of this exposure period, the temperature of the chamber shall be gradually decreased, within a 2-hour period, to 70° ± 5°F, and the relative humidity shall remain at 95 percent. The spin system shall remain in this chamber for a minimum period of 4 hours. At the end of this exposure period, the spin system shall be removed from the test chamber, and it shall comply with the requirements of 3.8.6. NOTE: The pins of the igniter assembly may be blown with dry compressed air after the spin system is removed from the test chamber.

4.5.5.7 Shock.- The spin system shall be securely mounted to a shock testing machine. The spin system shall be shocked three times, in both directions along each of three mutually perpendicular axes, for a total of 18 shocks. One of the three mutually perpendicular axes shall correspond to the longitudinal axis of the gas generator. The spin system shall comply with the requirements of 3.8.7 during this test.

4.5.5.8 Vibration.- The spin system shall be securely mounted to a vibration machine and subjected to the vibrations specified in 3.8.8. The spin system shall be subjected to these vibrations along each of two perpendicular axes which lie in the spin system plane. One of the axes shall correspond to the longitudinal axis of the gas generator. The spin system shall comply with the requirements of 3.8.8 during this test.

4.5.5.9 Spin rate (calculated).- The spin system shall be monitored for the parameters specified in the spin rate formula. The spin rate shall be calculated, by means of the following formula, to determine compliance with 3.8.9:

$$\text{Spin Rate} = \frac{(\text{average thrust})(\text{moment arm})(\text{action time}) \times 60}{2\pi \times (\text{moment of inertia})}$$

Moment Arm = 0.979 feet (11.75 inches).
Moment of Inertia = 9.0 slug-feet^2.
Average Thrust, lbs. = Average thrust measured during action time.
Action Time, seconds = Time interval as measured between points on pressure trace which are 10 percent of maximum operating pressure (the 10

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percent pressure points to be on the ascending and descending portion of trace). The maximum operating pressure shall be the highest pressure developed by the gas generator (excluding any pressure spike caused by ignition).

4.5.5.10 Weight.- The spin system shall be weighed on a Government-approved weighing device, readable to tenths of a pound, and shall comply with the requirements of 3.8.10.

5. PACKAGING

5.1 Spin system components (excluding gas generator assembly).

* 5.1.1 Preservation, packaging, packing, and marking.- Preservation, packaging, packing, and marking shall be in accordance with FED. STD. No. 356 or as specified in the contract or purchase order.

5.2 Gas generator assembly.

5.2.1 Preservation, packaging, and marking.- Preservation, packaging, and marking shall be in accordance with dwg. 8861961.

* 5.2.2 Packing and marking.- Packing and marking shall be in accordance with FED. STD. No. 356 or as specified in the contract or purchase order.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

* 6.1.1 Procurement requirements.- (a) Title, number, and date of this specification, (b) selection of methods of packaging or packing (see 5.1.1 and 5.2.2), and (c) first article sample requirements.

6.1.2 Data requirements.- Data specified in 3.4.2, and contractor engineering documentation specified in 4.4.4.1, will be delivered as identified on a numbered DD Form 1664 when specified on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence.- Records of the contractor's quality control and inspections which can be verified.

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* 6.3 Inspection of connectors.

6.3.1 Height of pins.- In the event a pin does not appear to be the same relative height of adjacent pins, a gage shall be used to determine the variation from normal height. Reject any connector that has a pin that varies by more than 0.040 inch maximum from the normal pin height of the individual connector. The normal height of the individual connector pins shall be determined by calculating the average height from measurements taken of pins adjacent to the pin that appears to be defective. The forward metal edge of the connector shall be utilized as the reference surface for all measurements.

6.4 Acceptance inspection equipment.- The contractor shall obtain approval of equipment designs, operating instructions, and calibration instructions from Commander, Picatinny Arsenal, Dover, New Jersey 07801, Attention: SARPA-QA-T.

6.5 Changes from previous issue.- The margins of this specification are marked with an asterisk to indicate where major changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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DOCUMENT IDENTIFIER AND TITLE

MIL-S-14767A Spin System, Atomic Warhead Section: M30.

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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